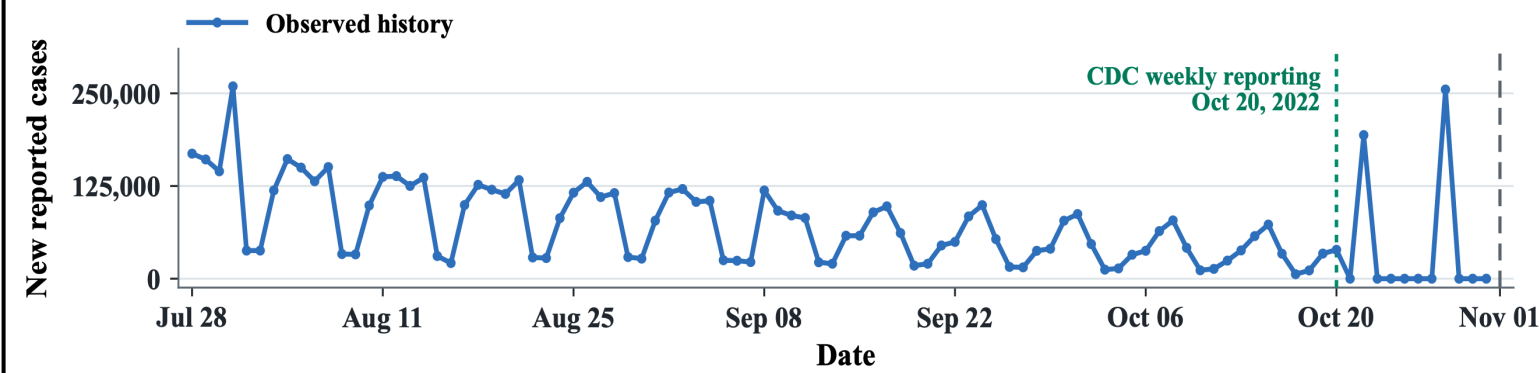


Reasoning Case Study: Why Context Preserves the Spike

US COVID-19 · Forecast horizon: Nov 1–6, 2022



EVENT

CDC switched from daily to weekly reporting — Oct 20, 2022

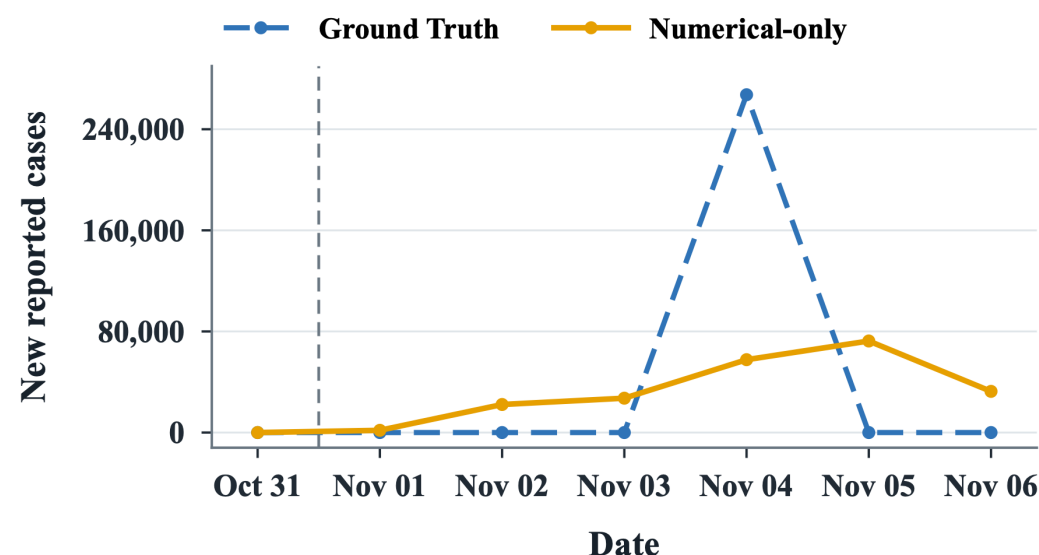
Numerical-only

- 1 **Finds the pre-break weekly cycle**
“Peaks: 4, 11, 18, 25, 32 = **7 days!** Yes exactly **period 7: index 4, 11, 18, 25, 32...** **Strong weekly seasonality with period 7.**”
- 2 **Recent cycles stop agreeing**
“Recent cycles: W6: i36-42: -0.581, -0.557,..., -0.7385 .**W7 vs W6 shifts high from p3 to p2. Not regular.**”
- 3 **The repeated floor remains ambiguous**
“Maybe there is a **regime shift to lower level around -0.7385.** Could be actual true if after index38 signal collapses? ...maybe data includes **missing values encoded as -0.7385**”
- 4 **A conservative compromise**
“**Conservative means avoid predicting outliers;** baseline may be lower than seasonal old but need include seasonal shoulder”
- 5 **Rejects sparse-spike replay; smooths the path**
“**This naive with spikes; not good.** [...] I think **transition-robust forecast** is one of best. [...] Let's make final maybe: {“forecast”: [-0.7303, -0.6353, -0.6121, -0.4704, -0.4016, -0.5870]}”

W / Context

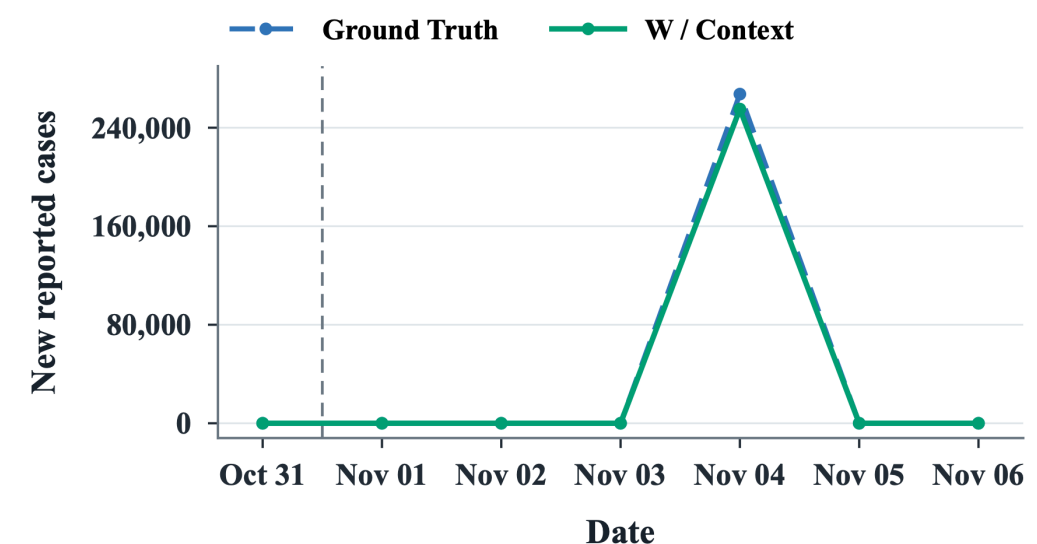
- 1 **The event becomes an explanatory hypothesis**
“CDC replaced daily updates with **weekly reporting, one record per week.** That explains flat period after Oct20?”
- 2 **The report-day hypothesis is tested**
“**Oct28 high could be weekly count;** but Oct22 high is Saturday unexplained.”
- 3 **Observed change, not the event alone, justifies persistence**
“**history shows regime change, use it.** Future report schedule is not an “event” supplied, but derived from change.”
- 4 **The last post-break calendar week becomes the template**
“E.g., Tue Nov1 use Oct25 Tue floor -> -0.7385; Wed use Oct26 floor; Thu use Oct27 floor; **Fri use Oct28 high 0.448;** Sat use Oct29 floor; Sun use Oct30 floor. That is **seasonal naive using calendar last week.**”
- 5 **No unsupported drift is added to the Friday spike**
“...after event there is a **structural break.** The first post-break full week Oct25-31 perhaps is the template under the **new regime;** there is no trend signal”

Smooths away the spike



MAE
60,970.9

Preserves the Friday spike



MAE
2,030.1
↓ 96.7%